POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Physics

Semester: Spring

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between damped oscillation and forced oscillation. Show that center of oscillation of compound pendulum lies beyond the center of gravity. 2+7

   b) A stretched string has a linear mass density of 5 gm/cm and a tension of 10N. A wave on the string has amplitude of 0.12 mm and a frequency of 100 Hz is travelling in negative x-direction. 4+2

      i. Write the wave equation with appropriate units.

      ii. At what average rate does the wave transport energy?

2. a) Define Newton's Ring and explain why center of Newton's Ring is dark. Also, determine the expression to find the unknown wavelength of monochromatic light using Newton's Ring apparatus. 2+7

   b) A class room has dimensions 20x15x5 m^3. The reverberation time is 3.5 sec. calculate the total absorption of its surfaces and the average absorption coefficient. 3+3

3. a) Define electric quardrupole. Hence determine the expression for electric potential due to quardrupole, at a point such that the point not lying along the axis of quardrupole. 1+8

   b) Light of wavelength 5000 A^0 is incident normally on a plane transmission grating. Find the difference in the angle of deviation in the first and third order spectra. The number of lines per cm on grating is 6000. 6

4. a) State Biot's and Savart's law. Find the expression of magnetic field strength due to a straight conductor which carries some current on it. (use Biot's and Savart's law) 2+7

   b) A parallel plate capacitor has a capacitance of 100×10^{-12} F, a plate of area of 100 cm^2 mica (k=7) is used as a dielectric, at 50 volts potential difference. Calculate the electric field intensity and magnitude of induced change. 6

5. a) Enlist the Maxwell's electromagnetic wave equation, with derivation in differential form. 9

   b) A wire of resistance 6 Ω is stretched by 20% what is the resistance of wire? Assume that resistivity remains same. 6

6. a) Derive the expression for schrodinger time dependent wave equation. 9

   b) If 10mH inductor and two capacitors of 5 μF and 2 μF are given, find the two resonant frequencies that can be obtained by connecting these elements in different ways. 6

7. Write short notes on: **(Any two)** 2×5

   a) Semi-conductor interms of Band theory.

   b) Optical fiber

   c) Lorentz force.